

REAL NUMBERS — MASTER CONTENT (CBSE + ICSE + PSEB)

For Class 9 + Class 10 (Unified, Board-wise differences included)

Perfect for: **Notes** | **Chapters Page** | **Concept Cards** | **MCQs** | **Solved Examples** | **PDF Export**

SECTION A — REAL NUMBERS (UNIFIED THEORY)

1. Introduction

Real Numbers = **Rational Numbers + Irrational Numbers** Represented on the **number line**, infinite, continuous.

Classification

- **Natural Numbers (N)**: 1, 2, 3, ...
- **Whole Numbers (W)**: 0, 1, 2, 3, ...
- **Integers (Z)**: ... -3, -2, -1, 0, 1, 2, 3 ...
- **Rational Numbers (Q)**: Numbers of the form $\frac{p}{q}$, $q \neq 0$
- **Irrational Numbers (Q')**: Non-terminating, non-repeating decimals
- **Real Numbers (R)**: $Q \cup Q'$

2. Euclid's Division Lemma (Class 10 – CBSE/ICSE/PSEB)

For any integers **a** and **b** ($b > 0$),

$$a = bq + r, 0 \leq r < b$$

Uses

- Finding **HCF**
- Proving irrationality
- Number system proofs

Example

Find HCF of 867 and 255. Apply lemma repeatedly $\rightarrow$ HCF = 51.

3. Fundamental Theorem of Arithmetic (FTA)

Every composite number can be uniquely expressed as a **product of primes**, except for order.

Applications

- Finding HCF/LCM using prime factorization
- Terminating/non-terminating decimal test
- Solving number theory problems

4. Irrational Numbers

Definition

Numbers that **cannot** be expressed as $\frac{p}{q}$.

Examples

- $\sqrt{2}, \sqrt{3}, \sqrt{5}$
- π
- Non-terminating, non-repeating decimals

Standard Proofs

- $\sqrt{2}$ is irrational
- $\sqrt{3}, \sqrt{5}$ irrational
- $p\sqrt{2}$ irrational if $p \neq 0$ rational

5. Decimal Expansions

Let $x = \frac{p}{q}$ in lowest form.

Case 1 — Terminating Decimal

If $q = 2^m \times 5^n$, then decimal terminates.

Case 2 — Non-terminating, Repeating

If q has **prime factors other than 2 or 5**.

Examples

- $7/8 \rightarrow$ terminating
- $1/3 \rightarrow$ repeating
- $13/40 \rightarrow$ terminating
- $22/7 \rightarrow$ repeating

6. Laws of Exponents for Real Numbers (Class 9 Focus)

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^{-n} = \frac{1}{a^n}$
- $a^{1/n} = \sqrt[n]{a}$

7. Surds & Indices (ICSE Emphasis)

Surd

An irrational root expression: $\sqrt{2}$, $\sqrt[3]{5}$, $\sqrt{7}$, $5\sqrt{3}$

Simplification Rules

- $\sqrt{ab} = \sqrt{a}\sqrt{b}$
- Rationalizing denominators
- Conjugates: $\frac{1}{a+\sqrt{b}}$

8. Number Line Representation

- Locating $\sqrt{2}$, $\sqrt{3}$ using Pythagoras
- Representing terminating decimals
- Representing repeating decimals

9. HCF & LCM (All Boards)

Methods

- Prime factorization
- Division method
- Euclid's algorithm

Relations

$$\text{HCF} \times \text{LCM} = \text{Product of numbers}$$

10. Board-wise Differences

CBSE Class 9

- Rational & irrational numbers
- Decimal expansions
- Exponents
- Number line
- Surds (light)

CBSE Class 10

- Euclid's Lemma
- FTA
- Irrationality proofs
- Decimal expansion test

ICSE Class 9

- Surds (heavy)
- Rationalization
- Laws of indices
- Real number operations

ICSE Class 10

- HCF/LCM
- Prime factorization

- Rationalization advanced
- Recurring decimals

PSEB Class 9

- Rational numbers
- Irrational numbers
- Exponents
- Surds basics

PSEB Class 10

- Euclid's Lemma
- FTA
- Decimal expansions
- HCF/LCM

SECTION B — EXAM-ORIENTED CONTENT

1. Important Formulas

- $a = bq + r$
- Terminating decimal test
- HCF–LCM relation
- Laws of exponents
- Surd simplification rules

2. Solved Examples (Mixed Boards)

Example 1 — Decimal Expansion

Check if $\frac{64}{1250}$ is terminating. $1250 = 2 \times 5^4 \rightarrow$ only 2 & 5 $\rightarrow$ **terminating**.

Example 2 — Irrationality Proof

Prove $\sqrt{5}$ is irrational. (Use contradiction method.)

Example 3 — HCF using Euclid

HCF(960, 1575) = 15.

Example 4 — Surd Rationalization (ICSE)

Rationalize

$$\frac{1}{\sqrt{3} - 1}$$

Multiply by conjugate →

$$\frac{\sqrt{3} + 1}{2}$$

SECTION C — MCQs (CBSE + ICSE + PSEB)

1. Which of the following is irrational?

(a) 0.333... (b) $22/7$ (c) $\sqrt{7}$ (d) 5.121212... **Answer: (c)**

2. If $2m \times 5n$ is denominator, decimal is

(a) repeating (b) terminating (c) irrational (d) none **Answer: (b)**

3. HCF(48, 180) =

(a) 6 (b) 12 (c) 24 (d) 36 **Answer: (b)**

SECTION D — SHORT NOTES FOR YOUR PORTAL

Concept Cards

- Real numbers
- Rational vs irrational
- Euclid's lemma
- FTA
- Decimal expansion test
- Surds
- Exponents
- HCF/LCM

Student-friendly summaries

- “Real numbers are all numbers on the number line.”
- “A decimal terminates only if denominator has 2 and 5.”
- “Every composite number has a unique prime factorization.”